Unit: 6 Unit 2: Exponents Radicals
Topic: Radicals

Name: ________________
Date: __________

Challenge (Habanero)

Q1. Simplify $\sqrt{64}$
(A) 8
(B) 16
(C) 32
(D) 4
(E) 2

Q2. Simplify $\sqrt[3]{27}$
(A) 3
(B) 9
(C) 27
(D) 1
(E) 2

Q3. A recipe calls for $\frac{1}{4}$ cup of sugar. If a chef wants to make 9 times the recipe, how much sugar will be needed in terms of $\sqrt{?}$?
(A) $\sqrt{9}$
(B) $\sqrt{16}$
(C) $\sqrt{25}$
(D) $\sqrt{36}$
(E) $\sqrt{49}$

Q4. A construction team needs to cut a beam into 5 equal parts. If the length of the beam is $\sqrt{225}$, how long will each part be?
(A) $\sqrt{5}$
(B) $\sqrt{9}$
(C) $\sqrt{15}$
(D) $\sqrt{45}$
(E) $\sqrt{25}$

Q5. Simplify $\sqrt{16x^2}$
(A) $4x$
(B) $4x^2$
(C) $2x$
(D) x^2
(E) $8x$

Q6. A baker is making a cake that requires a $\sqrt[3]{8}$-inch layer of frosting. How many $\frac{1}{2}$-inch layers can the baker make with 4 inches of frosting?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 12

Q7. Simplify $x^{\frac{1}{2}} \cdot x^{\frac{1}{2}}$
(A) $x^{\frac{1}{4}}$
(B) $x^{\frac{1}{2}}$
(C) x
(D) x^2
(E) x^3

Q8. A carpenter is building a staircase with $\sqrt{81}$ steps. If each step requires 5 planks of wood, how many planks will be needed in total?
(A) 25
(B) 40
(C) 45
(D) 50
(E) 405

Open Ended Questions (FRQ)

Q9. A chef is making a recipe that calls for $\frac{3}{4}$ cup of flour. If the chef wants to make $\frac{1}{3}$ of the recipe, how much flour will be needed in terms of $\sqrt{?}$. Show your work and simplify your answer.

Q10. A construction team needs to build a bridge that is $\sqrt[3]{64}$ meters long. If the team can build $\frac{1}{2}$ meter per hour, how many hours will it take to complete the bridge? Show your work and simplify your answer.

Chef's Tips

- Check for simplified radical form
- Watch for correct units
- Ensure students show work for free-response questions